MIL-M-36849(DM)
31 July 1964

MILITARY SPECIFICATION

MOUNTING BOARD, ELECTROCARDIOGRAPH RECORDS

1. SCOPE

1.1 This specification covers a device for mounting electrocardiograph tapes to yield a permanent record for inclusion into health records in Military medical facilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-00225/3	Aluminum Alloy Bar, Rod, and Wire, 2011.
QQ-A-00225/5	Aluminum Alloy Bar, Rod, and Wire (Rolled or Drawn), 2017.
QQ-A-00250/8	Aluminum Alloy, 5052, Plate and Sheet.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

Military

MIL-C-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Federal

Fed. Std. No. 66 Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

Defense Medical Supply Center

20746 Mounting Board, Electrocardiograph, Records.

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The mounting board shall be fabricated from aluminum alloy and corrosion resisting steel components as hereinafter specified and as shown on DMSC Drawing 20746.

3.1.1 Frame. The frame, side strip and fabricated angle brackets shall be 5052 aluminum alloy, temper H32 in accordance with Specification QQ-A-00250/8.

3.1.2 Roller. The roller shall be 2011 aluminum alloy, temper T3 in accordance with Specification QQ-A-00225/3.

3.1.3 Tape cutter. The tape cutter shall be AISI 303 corrosion resisting steel conforming to Standard 66.

3.1.4 Rivets. Rivets shall be 2017 aluminum alloy in accordance with Specification QQ-A-00225/5 of suitable temper to permit setting heads without cracking.

3.2 Style, design and dimensions. The mounting board shall consist of a frame, roller and tape cutter as shown on the drawing. The design shall be such as to permit mounting 12 to 14 electrocardiograph lead tapes, 4 inches long to make up a single complete patient record (when mounted in accordance with operating instructions specified in 3.6). Dimensions and tolerances shall be as shown on the drawing. Tolerances on frame and roller insofar as squareness and concentricity are concerned shall be held within acceptable limits of commercial practice to assure the proper functioning of the end item.

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3.3 Operation. The aluminum roller component shall function as the mandrel on which the 3 inch cardboard core of an 8 inch width library type tape roll (see 6.1) shall fit (the roll of tape shall not be included as part of the mounting board unit). When the roller with tape roll, mounted so that it feeds with the adhesive side up, is fitted into the frame (secured to a table top) and seats itself in the downward sloping blind slots, it shall be possible to pull the tape in its full width (adhesive side up) smoothly and evenly over the angulated frame surface with the scribed pattern. Uniformity of tape feed shall be attainable by aligning and evenly securing the adhesive tape end across its full width to the flat back of the tape cutter (ruler), pulling the ruler (with tape end secured), and at the same time exerting slight end pressure to retain the left edge of the cutter against the guide strip riveted to the left side of the frame.

3.4 Finish. The aluminum components only (frame and roller) shall be clear anodized in accordance with Specification MIL-C-8625 for type II anodic coating. The scribed or etched lines in the pattern on the top surface of the frame (as shown in View A-A on sheet 2 of the drawing) shall be filled with a suitable black marking material to yield a clear cut permanent design readily visible through a layer of the 8 inch width library tape (see 6.1). The corrosion resisting steel tape cutter shall be treated (passivated or electro-polished) to remove contamination and restore passivity of the base material.

3.5 Identification marking. Each mounting board shall be permanently marked on one side of the frame showing the serial number, the name or registered trademark of the manufacturer, the contract or purchase order number and the marking "U.S.". The markings shall be legible, permanent, and shall be accomplished by stamping or by the use of metal plate(s), permanently affixed.

3.6 Installation and operating instructions. The contractor shall enclose with each unit two copies each of an instruction brochure or booklet which shall describe the proper procedure for mounting the frame of the unit onto a wood or metal table top or other suitable working surface. The type, number, and size of hardware components required for mounting shall be clearly set forth. In addition, the following instructions for operation of the unit shall be included:

OPERATING INSTRUCTIONS FOR ECG RECORDS MOUNTING BOARD.

In addition to the mounting board, a cutting board is required to prepare tapes for mounting. Both items as well as the operator's hands should be clean before starting. To prepare the mounting board, a roll of library tape as indicated in 6.1 should be loaded onto the roller and the roller inserted into the blind slots on the frame, so that the tape will peel off the roll with the adhesive side up. The tape roll should be positioned on the roller so that when it is drawn over the top surface of the frame, the tape edges will coincide as closely as possible with the vertical lines scribed on the extreme left and right sides of the top surface of the frame. To draw the tape down evenly over the frame, the tape end should be aligned and secured to the flat back of the tape cutter (ruler) and pulled, while exerting slight downward pressure on the top surface of the frame and sufficient end pressure on the cutter to hold it in

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contact with the guide strip on the left side of the frame. The tape should be drawn down over the top of the frame to the point where the bottom edge of the ruler is flush with the lower end of the top surface. The tape cutter should then be removed in a single quick snap stroke. Electrostatic attraction will hold the adhesive backing tape taut while the individual lead tapes are mounted. With the mounting board prepared as described above, the electrocardiograph (ECG) tapes to be mounted should be prepared in the following manner:

(a) Draw the ECG tape (as it is delivered from the electrocardiograph) across the cutting board, beginning with the last lead taken.

(b) Cut tapes in reverse order, starting from the last lead taken and working toward the first, for both limb and chest leads. While cutting the lead tapes, observe the height and position of the complex on each. In technically good tracings, the complex does not over-run or approach either the upper or lower margins more closely than 5 mm. Identify and hold aside any ECG tapes which do not meet this standard and note the comments in (d) below for handling such tapes. The left edge of each tape should be carefully trimmed so that is is square and cut 5 mm. (5 small spaces on the cross hatched portion of the tape) from the left side of the standardization signal at the base line. When the cutting of all tapes is completed, the limb and chest lead tapes should be arranged in the order taken alongside the cutting board (as a result of cutting tapes in reverse order).

(c) Arrange limb and chest lead tapes into two stacks on the cutting board with the left, top and bottom edges lined up. Carefully cut each stack (on the right end) to a length of 4 inches as accurately as possible.

(d) If the complex on the ECG tape is of good technical quality, trim off the top and bottom margins plus 5 mm. of the adjacent cross hatched portion on each tape, leaving only the cross hatched pattern and the complex trace. This will reduce the width of the tape to 40 mm. If the complex should come close to (within 5 mm.) or overrun one of the margins, that margin should not be trimmed off. Instead, the opposite margin plus 10 mm. of the adjacent cross hatched portion should be trimmed to yield a tape 40 mm. wide.

(e) Mount the trimmed ECG tapes for the limb leads on the left and the chest leads on the right side of the adhesive tape surface drawn across the top of the mounting board frame. The scribed lines on the frame surface will be visible through the adhesive tape and will outline a pattern of 14 areas 40 mm. wide by 4 inches long. Start with the ECG tape for the first limb lead. Align the top edge of the ECG tape with the upper horizontal line scribed on frame and with left edges of both the ECG and adhesive backing tapes flush, wipe the ECG tape uniformly into contact with the backing tape surface, being careful

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not to scuff or mar the sensitive surface of the ECG tape. In similar fashion using lines scribed on the frame as a guide, mount the balance of the ECG limb tapes, working downward in order on the left side of the 8 inch adhesive backing tape drawn over the mounting board frame. In mounting the tapes for the chest leads on the right side of the mounting board, the left ends of these ECG tapes should abut and line up with the ECG lead tapes already mounted on the left side. Otherwise, the procedure for mounting the chest lead tapes is the same as for mounting the limb lead tapes.

f) With all ECG tapes mounted on the adhesive library tape backing, draw the assembled record downward to expose 2-1/2 to 3 inches of the adhesive backing tape above the two top ECG lead tapes. With the tape cutter firmly positioned on the frame so that its left end bears against the frame guide strip and its lower cutting edge is one inch above the top of the first row of ECG lead tapes, shear the tape record free by pulling it in a single rapid motion. The tape cutter will then be secured to the end of the library tape ready for pulling down in preparation for mounting the next series of lead tapes. The one inch exposed strip of adhesive backing projecting above the mounted record is used to secure the form SF520 which becomes page 1 of the complete ECG record. However, before the mounted tape record is assembled to the form, it should be examined and trimmed flush on the cutting board to remove any ECG tape edges overhanging the adhesive backing. The ECG tape record is completed when the form SF520 is aligned over the tape record and pressed into contact with the 1 inch adhesive strip projecting at the top.

Notes

(1) Usually, only 12 ECG lead tapes will be required for a complete record. However, the number and manner of arranging lead tapes as described above is subject to change at the discretion of the physician in charge.

(2) It is suggested that the 40 mm. and 4 inch cutting distances referenced above be clearly identified on the cutting board by accurately marking these distances on the board parallel to the cutting edge with a thin strip of colored tape or marking crayon. This will help to speed up the tape cutting operation.

3.7 Workmanship. Workmanship shall be first class throughout. The mounting board shall be free from defects which detract from its appearance or may impair its serviceability. Rivets shall be squarely and uniformly set.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105.

4.3 Examination. The mounting board shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) mounting board, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table 1. Packaging Quantities

Unit package	Exterior container
1 unit	4 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spillage under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type I, class 2.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of Specification PPP-B-636. In addition, all seams, including manufacturer's joint, shall be closed by tape conforming to the requirements of Specification PPP-T-76. The tape shall be not less than 2 inches wide. Three strips shall be applied to both top and bottom so that all seams and joints shall be covered their full length. Tape applied over the lengthwise seams shall extend a minimum of 2 inches onto each of the end panels. Tape applied over the end edge seams shall extend a minimum of 2 inches along the side and top, and side and bottom panels.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

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NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Specification PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table 1; or PPP-B-636, type I, class 1.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129. In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated.

6. NOTES

6.1 The library backing tape referenced in 3.3 and 3.4 is No. 810 library tape, matte finish, put up in rolls 8 inches wide by 72 inches long on a cardboard core, 3 inches in nominal inside diameter obtainable from Minnesota Mining and Manufacturing Company.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet military requirements.

Custodians:
Army. - MD
Navy. - BuMed
Air Force. - AFMSHB(03)

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - BuMed
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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-M-36249(DM), MOUNTING BOARD, ELECTROCARDIOGRAPH RECORDS dt

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

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